

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY**

**JNANASANGAMA, BELAGAVI - 590018**



**MINI PROJECT**

**DISSERTATION**

**on**

**“INTEGRATED TEMPERATURE-BASED IRRIGATION AND ANIMAL INTRUSION  
PREVENTION USING ESP32”**

***Submitted in partial fulfilment for the award of degree of***

**Bachelor of Engineering**

**in**

**Electronics and Communication Engineering**

***Submitted by***

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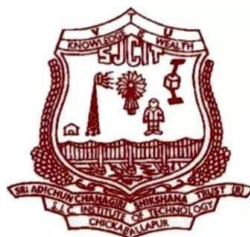
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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

**SJC INSTITUTE OF TECHNOLOGY, CHICKABALLAPUR-562101**

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### **CERTIFICATE**

Certified that the project work entitled **“INTEGRATED TEMPERATURE-BASED IRRIGATION AND ANIMAL INTRUSION PREVENTION USING ESP32”** carried out by **Mr.Yashwanth N(1SJ23EC188), Mr.Uday Chandra S(1SJ23EC175), Mr.Punith Kumar R(1SJ24EC411)** a bonafide student of SJC Institute of Technology, in partial fulfillment for the award of **BACHELOR OF ENGINEERING IN ELECTRONICS AND COMMUNICATION ENGINEERING** of **Visvesvaraya Technological University, Belgaum** during the academic year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of Guide

**Dr. Bhaskar S**

Signature of HOD

**Dr. C Rangaswamy**

## DECLARATION

We Yashwanth N, Uday Chandra S, Punith Kumar R student of V semester **B.E in Electronics and Communication Engineering** at **SJC Institute of Technology**, Chickballapur, hereby declare that the Mini Project entitled “**INTEGRATED TEMPERATURE-BASED IRRIGATION AND ANIMAL INTRUSION PREVENTION USING ESP32**” embodies report of our project work carried out independently by me, under the guidance of **Dr Bhaskar, Professor, Department of ECE, SJCIT, Chickballapur** as partial fulfillment of requirement for the award of **B.E Degree** during year **2025-26**. Further, the matter embodied in the dissertation has not been submitted previously by anybody for the award of the award of any degree to any other university.

Place: Chickballapur

Date: 05-12-2025

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Submitting devout pranamas and seeking the blessings of his holiness Paramapoojya Jagadguru Padmabhushana Sri Sri Sri Dr. Nirmalanandanatha Mahaswamiji and Poojya Sri Sri Mangalanath Swamiji.

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## CHAPTER-1

### INTRODUCTION

Agriculture remains the backbone of many developing economies, yet it continues to face persistent challenges related to water management and crop protection. In traditional farming practices, irrigation is often performed manually without considering real-time variations in soil moisture or environmental conditions. This leads to issues such as over-irrigation, which wastes water and damages plant roots, or under-irrigation, which results in reduced crop growth and yield. With increasing water scarcity and the rising need for sustainable farming, the adoption of intelligent irrigation systems has become essential.

Another major challenge faced by farmers is crop damage caused by stray animals, such as cattle, wild boars, and other intruding species. These animals enter farmlands in search of food and frequently destroy crops, causing significant financial losses. Conventional methods of securing farmland, such as physical fencing or manual surveillance, are often ineffective, labor-intensive, and costly to maintain. Therefore, there is a strong need for an automated, real-time intrusion detection mechanism that can alert farmers and deter animals efficiently.

To address these dual challenges, this project proposes an Integrated Temperature-Based Irrigation and Animal Intrusion Prevention System using the ESP8266 NodeMCU microcontroller. The ESP8266, known for its low power consumption and built-in Wi-Fi capability, serves as the central processing unit, enabling seamless monitoring and automation.

The system uses a soil moisture sensor to continuously measure soil water content. When moisture levels drop below a predefined threshold, the ESP8266 activates a relay-controlled water pump, ensuring that irrigation occurs only when needed. Temperature-based analysis can further optimize irrigation by adjusting watering cycles in relation to environmental conditions.

On the security side, an ultrasonic sensor detects objects or animals approaching the crop boundary by measuring distance variations, while a PIR sensor identifies movement within the protected zone. When intrusion is detected, a buzzer is triggered to scare away animals and notify nearby farmers. This automated alert mechanism minimizes crop damage and reduces the need for constant human supervision.

The system is powered by a rechargeable battery, ensuring continuous operation even in remote areas with limited access to electricity. Compact, low-cost, and easy to deploy, the proposed integrated solution aligns with the principles of smart farming and Internet of Things (IoT) technology.

By combining irrigation automation with intrusion prevention in a single platform, this project offers a practical and efficient approach to improving agricultural productivity, conserving water resources, and safeguarding crops. The solution demonstrates how modern IoT-based technologies can transform traditional farming into a more intelligent, sustainable, and resilient system.

## 1.1 Problem Statement

Agricultural fields often suffer from two major issues: inefficient water usage due to manual irrigation practices and frequent crop damage caused by stray animals entering the farmland. Traditional irrigation methods do not consider real-time soil moisture or environmental conditions, leading to over-irrigation or under-irrigation, which affects crop health and reduces productivity. Additionally, farmers lack automated systems to detect and deter animal intrusion, resulting in significant economic losses. There is a need for a low-cost, automated, and integrated system that can intelligently monitor soil conditions, manage irrigation, and simultaneously detect and prevent intrusion using reliable sensing technology. This project aims to design and implement an IoT-enabled solution using ESP8266 NodeMCU that addresses both irrigation inefficiencies and farm security challenges in a unified system.

## 1.2 Objectives

- To design an automated irrigation system that monitors soil moisture levels in real time and activates the water pump only when the soil becomes dry.
- To integrate temperature-based decision-making to improve irrigation efficiency under varying environmental conditions.
- To develop an animal intrusion detection mechanism using ultrasonic and PIR sensors to identify the presence or movement of animals near the field.
- To implement an alert system using a buzzer to deter animals and notify intrusion instantly.
- To utilize the ESP8266 NodeMCU as a low-cost, Wi-Fi-enabled microcontroller for efficient sensor data processing and system control.
- To ensure continuous system operation using a rechargeable battery for remote agricultural environments.
- To create an integrated, scalable, and energy-efficient solution combining irrigation automation and farm security in a single smart IoT-based system.

## CHAPTER 2

### LITERATURE SURVEY

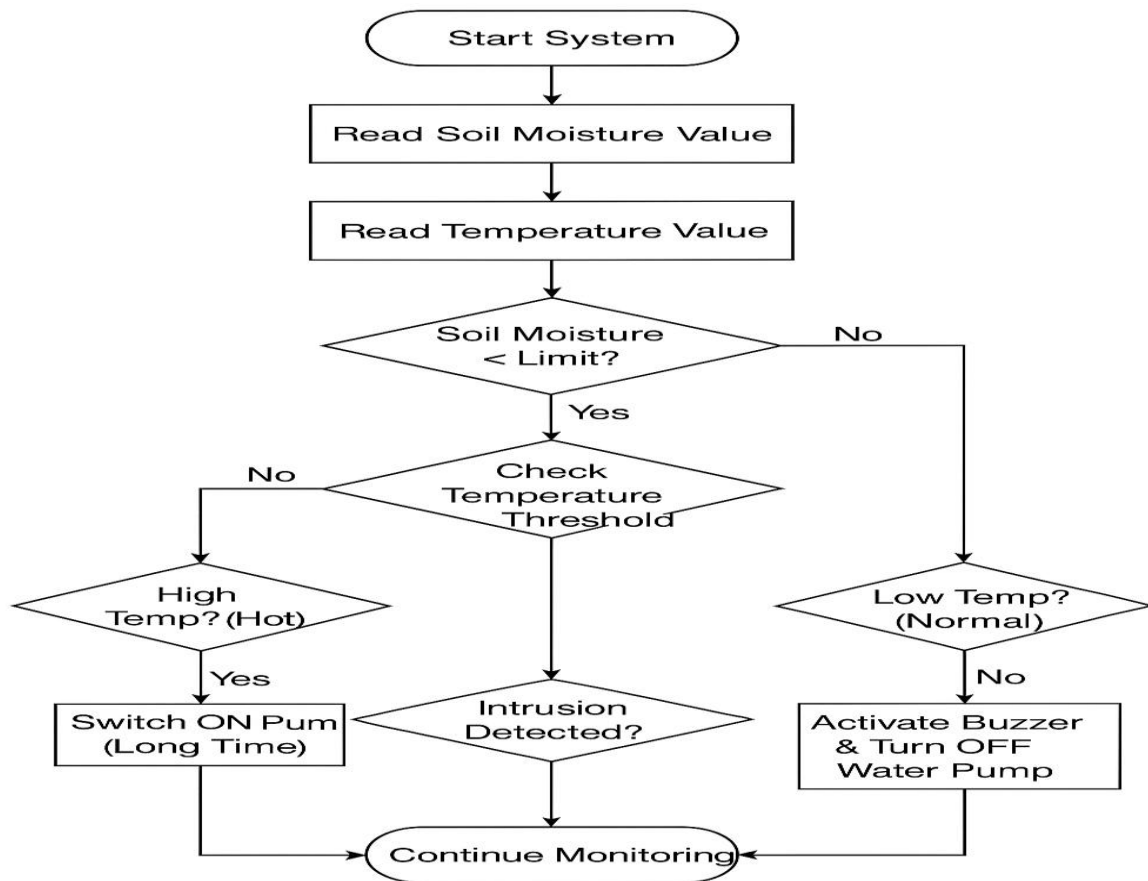
- 1 Khalid and Radzi [1] designed a **solar photovoltaic-powered IoT-based farm protection system** capable of detecting human and animal intrusions. Their system integrates PIR sensors and ultrasonic sensing with a Wi-Fi communication module to alert farmers in real time. The use of solar energy enhances sustainability and makes the model suitable for remote areas. Although effective for surveillance, the system does not integrate irrigation automation, leaving scope for a more comprehensive smart farming solution.
- 2 Gowda et al. [2] presented an IoT-based **smart agriculture framework** that focuses on environmental monitoring and automated control of irrigation. Their work utilizes sensors to measure soil moisture, temperature, and humidity, enabling real-time precision farming. However, the study primarily emphasizes irrigation and crop monitoring and does not address farm security issues such as animal intrusion.
- 3 Kumar et al. [3] developed a farm protection and monitoring system using an **ESP32 with a camera module** to capture real-time images of intruders. The system effectively identifies unauthorized human or animal presence and sends alerts to the farmer. While the camera-based approach enhances accuracy, the system has higher power consumption and processing overhead, making it less suitable for small-scale farms or low-power environments.
- 4 Nayyar and Puri [4] proposed an IoT-based agricultural stick embedded with **smart sensors** for continuous monitoring of temperature and soil moisture. Their design provides mobility and ease of deployment in various parts of the field. Although innovative, this solution focuses on environmental sensing rather than intrusion prevention or integrated pump control.
- 5 A comprehensive review by Yazdinejad et al. [5] examined various **security aspects of smart farming**, including cyber threats, sensor vulnerabilities, and data security challenges. The study emphasizes the importance of designing secure IoT architectures for agriculture due to the increasing reliance on connected systems. The review highlights strong motivations for building secure and resilient IoT-based farm protection systems such as the one proposed in this project.

- 6 Subramanyam et al. [6] developed an IoT-based protection system using **ESP32 and camera modules** for preventing crop damage from animals and human theft. Their system sends real-time images and alerts, making it suitable for surveillance. However, like earlier camera-based solutions, the design lacks irrigation automation and relies heavily on network bandwidth for image transmission.
- 7 Chavan and Thakur [7] proposed a **PIR sensor-based farm protection system** that sends GSM alerts to farmers during intrusion events. Their model effectively provides real-time notification and simple deployment. However, it lacks advanced sensing integration and does not incorporate automated irrigation or multi-sensor decision-making, limiting its scope.

## CHAPTER-3

### METHODOLOGY

#### 3.1 FLOWCHART



#### Flowchart Explanation

##### 1. Start the System

The system initializes all sensors and actuators including soil moisture sensor, temperature sensor, ultrasonic sensor, PIR sensor, relay module, and buzzer.

##### 2. Read Soil Moisture Value

The ESP8266 reads analog signals from the soil moisture sensor to determine soil conditions such as dry, moist, or wet.

3. Read Temperature Value The ESP8266 reads ambient temperature to decide irrigation duration. High temperature requires more watering, while normal temperature requires shorter watering.

#### 4. Check Soil Moisture Threshold

If the soil is dry, the system proceeds to irrigation logic. If the soil is moist or wet, the pump remains off and the system moves to intrusion detection.

#### 5. Temperature-Based Irrigation Decision

If the soil is dry and temperature is high, the pump runs longer. If the temperature is normal, the pump runs for a shorter time.

#### 6. Intrusion Detection (Parallel Process)

Ultrasonic and PIR sensors continuously check for animal movement or presence near the field.

#### 7. Intrusion Detected?

If both ultrasonic and PIR sensors detect activity, intrusion is confirmed.

#### 8. If Intrusion is Detected

The buzzer turns on to scare animals and the water pump turns off to prevent accidents or water wastage.

#### 9. Continue Monitoring

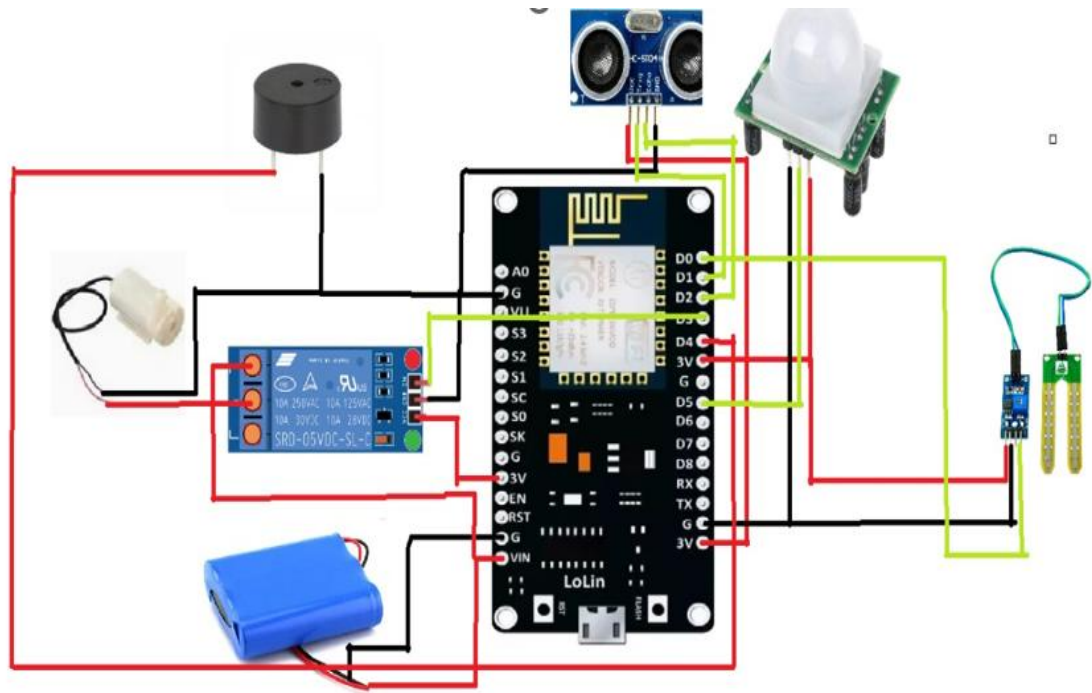
When intrusion ends, the buzzer turns off and the system resumes moisture and temperature monitoring to continue the irrigation cycle.

#### Summary of System Behaviour

| Condition              | Action                  |
|------------------------|-------------------------|
| Soil Dry + Normal Temp | Pump ON (short time)    |
| Soil Dry + High Temp   | Pump ON (long time)     |
| Soil Moist/Wet         | Pump OFF                |
| Intrusion Detected     | Pump OFF + Buzzer ON    |
| No Intrusion           | Normal irrigation cycle |

### 3.2 HARDWARE REQUIREMENTS

#### Circuit diagram



This circuit schematic shows a complete smart automation system that uses the NodeMCU (ESP8266) as its main controller. The NodeMCU can read data and control different parts because all of the sensors and output devices are connected to it. The NodeMCU is connected to the soil moisture sensor, which can tell how wet the soil is. The sensor's output pin is attached to a digital pin.

The battery and the water pump are both connected to the switch. The NodeMCU turns on the relay, which sends power from the battery to the pump, which sends water to the plant. The board is connected to a PIR motion sensor that detects movement. When it does, the buzzer goes off to let you know. There is also an ultrasonic sensor (HC-SR04) that is connected to the NodeMCU by trigger and echo pins. You may use this sensor to keep an eye on the water level in a tank or find impediments. It can also measure distance. A digital pin powers the buzzer, which can make noise whenever the microcontroller sends a signal. A rechargeable battery powers the whole circuit. The positive terminal is connected to the Node MCU's VIN pin, and all the parts share the same ground. This system can measure distance, sense motion, monitor soil moisture, pump water, and sound alarms.

## COMPONENTS REQUIRED

### 1.Ultrasonic sensor



An ultrasonic sensor is a piece of electronics that uses sound waves to measure distance. The sensor sends out high-frequency ultrasonic waves (typically 40 kHz) from its transmitter. These waves travel through the air, hit an object, and then bounce back to the sensor's receiver. The sensor measures how long it takes for the waves to come back. Then it uses that time to figure out how far away the object is. The HC-SR04 is a highly common ultrasonic sensor with four pins: VCC, GND, TRIG, and ECHO. Robots often use them to find obstacles, automobiles use them to park, water-level monitors use them, and many other automation projects use them. These sensors operate well inside, are cheap, and don't need light to work, so they may be used even when it's completely dark. They can't, however, reliably detect very delicate objects, and their range is only a few meters. In general, ultrasonic sensors are good tools for measuring things that are close together in a lot of electronics and IoT applications.

### 2.Soil Moisture Sensor



The Soil Moisture Sensor tells you how much water is in the soil. It works by measuring the soil's electrical resistance or capacitance. The sensor delivers a signal to start watering

when the soil moisture level drops below a certain point. The ESP32 gets this information and tells the user to water the plants. This sensor makes sure that the soil is always the right amount of moist for plants to grow. It's easy to use and works well with a lot of different kinds of soil.

### 3.PIR Sensor (Motion Detection Sensor)



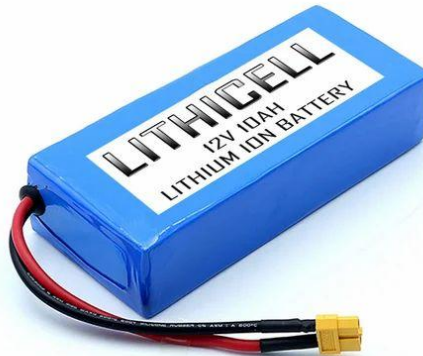
The PIR (Passive Infrared) Sensor picks up on motion or movement in the area. It works by using the infrared light that people and animals naturally give off. The PIR sensor gives out a signal when it detects a change in infrared radiation when something moves within its detection range. The proposed system employs a PIR sensor to identify illicit movement within the agricultural zone. The sensor picks up movement in the crop area and sends an alert signal to the ESP32 CAM module if an animal or burglar gets in. The ESP32 CAM can take a picture or video of the field and let the user know when it gets the signal. This helps keep crops safe, cut down on theft, and keep fields safe, especially at night. The PIR sensor is great for outdoor use because it doesn't require much electricity and can sense a vast area.

### 4.Buzzer



The buzzer is a device that lets you know when an alarm is going off. The buzzer makes a sound when a parameter, like temperature, soil moisture, or water pressure, goes above the safe limit.

## 5. Rechargeable Battery



The whole system gets its power from a rechargeable battery. The system is designed to work well with battery backup because agricultural fields may be far away from a stable electrical source. You may also connect the battery to a solar charging station to make it last longer and be more environmentally friendly. This means that the technology can be used in the field in real time.

## 6. Relay Module



The relay module turns on and off things like lights and water pumps on the farm. It works like an automatic switch that listens to commands from the ESP32 controller. The relay turns on the pump when the soil moisture level drops below a certain point or when irrigation is needed. The pump turns off when the moisture level goes back to normal.

## 7. Water pump



A water pump is a machine that transports water from one place to another. By putting pressure on pipes, it lets water flow quickly. People use water pumps a lot in their homes to move water from wells or storage tanks to above tanks. Also, they are used in agriculture for irrigation, in industry for cooling systems, and in water treatment plants. There are three main types of water pumps: centrifugal pumps, submersible pumps, and motor pumps. Each type is made for a different need. Submersible pumps work underwater and are best for deep wells. Centrifugal pumps work on the surface and are best for regular pumping jobs. Water pumps are very important because they keep a steady flow of water for both everyday tasks and big projects.

## 8. ESP8266 Node MCU



The ESP8266 NodeMCU is a cheap microcontroller with WiFi that is made for Internet of Things (IoT) projects. It has a powerful 32-bit processor, WiFi built in, and a number of GPIO ports that may be used to connect sensors, modules, and actuators. The Arduino IDE or Lua can be used to program the board, which is great for people who are just starting out. You can easily control devices, send data to cloud platforms, and make smart

home systems with apps like Blynk because it has WiFi. The Node MCU runs on 3.3V and is commonly used in projects like home automation, sensor monitoring, water level systems, and wireless data logging. It is one of the most popular boards for IoT development since it is cheap, flexible, and easy to use.

Another major advantage of this system is low cost compared to CCTV-based surveillance systems. Traditional surveillance systems require high storage capacity, external power supply, and heavy wiring, making them expensive and complex to maintain. In contrast, this IoT-based system is compact, energy efficient, and much more affordable for small and medium scale farmers. The entire setup can be deployed in both small farms and large orchards, making it versatile .

The system also encourages farmers to adopt data-driven agriculture. Over time, by observing the recorded temperature, humidity, and soil moisture readings, farmers gain a better understanding of crop water requirements. This knowledge helps them plan irrigation schedules more effectively in the future. If cloud storage is integrated later, this data could also be used with machine learning algorithms to predict and automate irrigation cycles.

In large-scale implementation, several ESP32-CAM units and sensors can be placed at multiple points across the field. All of them can send updates to the same Telegram bot. This allows distributed monitoring across the whole farmland and supports the concept of precision agriculture, where each portion of the field is monitored individually for accurate decision-making.

In summary, the system integrates security, irrigation management, climate monitoring, and wireless communication into one unified solution. It is practical, affordable, and can be adopted easily by farmers with basic training. The use of IoT and renewable energy also aligns with future advancements in Indian agriculture, where traditional practices are gradually being enhanced with technological support.

Another important aspect to consider is how this system changes the daily routine and workload of the farmer. Traditionally, a large amount of time is spent manually inspecting fields, checking soil condition, observing plant growth, and protecting crops. This physical effort increases even more during crop maturity stages when the plants are more vulnerable to damage from animals and climatic variations. With this automated setup,

the farmer's involvement shifts from continuous physical monitoring to smart supervision. The system also improves resource efficiency. By using sensors to measure exact conditions, the farmer uses water, fertilizers, and labour only when necessary.

## **CONNECTION**

### **HARDWARE CONNECTIONS (NODEMCU ESP8266 + SENSORS + RELAY + BUZZER + PUMP)**

#### **1. Soil Moisture Sensor (Digital Output Type)**

Soil Sensor → NodeMCU

- VCC → 3.3V or 5V (recommended 3.3V)
- GND → GND
- DO → D0

#### **2. Ultrasonic Sensor (HC-SR04)**

The ultrasonic sensor needs 5V. It still works with 3.3V logic on ECHO because HC-SR04 gives 5V output → must use a voltage divider.

Ultrasonic → NodeMCU

- VCC → 5V
- GND → GND
- TRIG → D1
- ECHO → D2 (through 5V→3.3V voltage divider)

#### **3. Buzzer**

Buzzer → NodeMCU

- + → D4
- – → GND

#### **4. PIR Motion Sensor (HC-SR501)**

PIR → NodeMCU

- VCC → 5V
- GND → GND
- OUT → D5

#### **5. Relay Module (Water Pump Control)**

Relay Input → NodeMCU

- IN → D3
- VCC → 5V
- GND → GND

Relay Output → Pump

pump uses 12V:

Use a separate 12V battery for pump

NodeMCU uses USB or 5V supply

Common ground MUST be connected.

#### **6. Power Supply**

You can use:

- NodeMCU via USB 5V
- Sensors powered by 5V
- Pump powered by 12V battery or 5V motor  
(through relay only)

## Complete Wiring Summary

| Component       | Signal Pin | Connects To |
|-----------------|------------|-------------|
| Soil Moisture   | D0         | D0          |
| Ultrasonic TRIG | TRIG       | D1          |
| Ultrasonic ECHO | ECHO       | D2          |
| Relay           | IN         | B3          |
| Buzzer          | +          | D4          |
| PIR Sensor      | OUT        | D5          |
| LED             | +          | D6          |
| All VCC         | VCC        | 5V          |
| All GND         | GND        | GND         |

## 3.3 SOFTWARE REQUIREMENTS

### 1. Arduino IDE

- Version 1.8.x or 2.x
- Used to write, compile, and upload the code to NodeMCU ESP8266.

### 2. Board Package

To support ESP8266 board in Arduino IDE:

- ESP8266 Boards Package

- Install  
Arduino IDE → File → Preferences → Additional Board Manager URLs:  
  
[http://arduino.esp8266.com/stable/package\\_esp8266com\\_index.json](http://arduino.esp8266.com/stable/package_esp8266com_index.json)
- Then  
Tools → Board → Boards Manager → Search “ESP8266” → Install

### 3. Required Libraries

- Your current code does not use DHT, LCD, Soil analog libraries, so only these are required:
- ESP8266 Core Libraries
- Installed automatically with board package.

### 4. Serial Monitor

- Baud rate: 9600
- Used to debug:
  - PIR motion status
  - Ultrasonic distance
  - Soil sensor output
  - Pump relay state

### 5. Operating System Compatibility

- Windows 7/8/10/11
- Linux
- macOS

### 6. USB Driver

Depending on your NodeMCU version:

If your board uses CH340:

- Install CH340 USB-to-Serial Driver

If your board uses CP2102:

- Install CP2102 Driver

**CODE:**

```
int water;

int trigpin=D1;

int echopin=D2;

long duration;

int distance;

int BUZZER=D4;

int pirPin = D5;

int ledPin = D6;

int buzzerPin = D4;

int relayPin =D3;

int pirState = HIGH;

int val = 1;

//Blynk Virtual Pins

#define VPIN_SOIL V0

#define VPIN_DISTANCE V1

#define VPIN_PIR V2

#define VPIN_PUMP_CONTROL V3

#define VPIN_BUZZER V4

// Blynk Function to Control Pump Manually

/*BLYNK_WRITE(VPIN_PUMP_CONTROL)

{
```

```
int pinValue = param.asInt(); // Get value from app button

digitalWrite(relayPin, pinValue);

Serial.print("Pump manually set to: ");

Serial.println(pinValue ? "ON" : "OFF");

}*/

void setup() {

  pinMode(D3,OUTPUT);

  pinMode(D0,INPUT);

  pinMode(trigpin,OUTPUT);

  pinMode(echopin,INPUT);

  pinMode(BUZZER,OUTPUT);

  pinMode(pirPin, INPUT);

  pinMode(ledPin, OUTPUT);

  pinMode(buzzerPin, OUTPUT);

  Serial.println("PIR Motion Sensor Test Starting...");

  Serial.begin(9600);

}

void loop() {

  water = digitalRead(D0);

  if(water == HIGH)

  {

    digitalWrite(D3,LOW);

  }

  else
```

```
{  
  
    digitalWrite(D3,HIGH);  
  
}  
  
    delay(400);  
  
    digitalWrite(trigpin,LOW);  
  
    delayMicroseconds(2);  
  
    digitalWrite(trigpin,HIGH);  
  
    delayMicroseconds(20);  
  
    digitalWrite(trigpin,LOW);  
  
    duration=pulseIn(echopin,HIGH);  
  
    distance=duration*0.034/2;  
  
    Serial.print("DISTANCE IN CM");  
  
    Serial.println(distance);  
  
    if ((distance<=100))  
    {  
  
        digitalWrite(BUZZER,HIGH);  
  
        Serial.println("OBJECT IS DETECTED");  
  
    }  
  
    else  
  
    {  
  
        digitalWrite(BUZZER,LOW);  
  
        Serial.println("OBJECT IS NOT DETECTED");  
  
    }  
  
}
```

```
val = digitalRead(pirPin);

if (val == HIGH)

{

    digitalWrite(ledPin, HIGH);

    delay(1000);

    if (pirState == LOW)

    {

        Serial.println("△ Motion detected!");

        pirState = HIGH;

        delay(1000);

    }

}

else

{

    digitalWrite(ledPin, LOW);

}

if (pirState == HIGH)

{

    Serial.println("Motion ended.");

    pirState = LOW;

}

}
```

## COMPLETE CODE EXPLANATION

### 1. Variable & Pin Declarations

```
int water;  
  
int trigpin = D1;  
  
int echopin = D2;  
  
long duration;  
  
int distance;  
  
int BUZZER = D4;  
  
int pirPin = D5;  
  
int ledPin = D6;  
  
int buzzerPin = D4;  
  
int relayPin = D3;
```

#### Meaning:

- water → reads soil moisture sensor digital output.
- **Ultrasonic Sensor:**
  - trigpin → sends trigger pulse.
  - echopin → receives echo pulse.
  - duration → time taken by echo pulse.
  - distance → calculated distance in cm.
- BUZZER and buzzerPin (both D4) → buzzer pin for alert.
- pirPin → input pin for PIR motion sensor.
- ledPin → LED indication when motion is detected.
- relayPin → controls the water pump through relay module.

## 2. PIR Motion Sensor Variables

```
int pirState = HIGH;
```

```
int val = 1;
```

### Meaning:

- val stores PIR digital reading (HIGH or LOW).
- pirState avoids repeated printing of "Motion detected" messages.

## 3. setup() Function – Runs Once at Startup

```
pinMode(D3,OUTPUT);
```

```
pinMode(D0,INPUT);
```

```
pinMode(trigpin,OUTPUT);
```

```
pinMode(echopin,INPUT);
```

```
pinMode(BUZZER,OUTPUT);
```

```
pinMode(pirPin, INPUT);
```

```
pinMode(ledPin, OUTPUT);
```

```
pinMode(buzzerPin, OUTPUT);
```

```
Serial.println("PIR Motion Sensor Test Starting...");
```

```
Serial.begin(9600);
```

### Meaning:

- Relay → OUTPUT
- Soil moisture → INPUT
- Ultrasonic TRIG → OUTPUT
- Ultrasonic ECHO → INPUT
- PIR → INPUT
- LED → OUTPUT
- Buzzer → OUTPUT
- Starts serial communication at 9600 baud rate

## 4. loop() Function – Runs Continuously

### A) SOIL MOISTURE SENSOR (D0)

```
water = digitalRead(D0);
```

```
if(water == HIGH)
```

```
{
```

```
    digitalWrite(D3,LOW);
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(D3,HIGH);
```

```
}
```

```
delay(400);
```

#### Meaning:

- If D0 == HIGH → Soil is **WET**
  - Relay OFF → Pump OFF
- If D0 == LOW → Soil is **DRY**
  - Relay ON → Pump ON

Relay logic:

- **LOW → OFF**
- **HIGH → ON**

Pump works automatically based on soil condition.

### B) ULTRASONIC SENSOR – Object/Animal Detection

#### Trigger Pulse

```
digitalWrite(trigpin,LOW);
```

```
delayMicroseconds(2);
```

```
digitalWrite(trigpin,HIGH);
```

```
delayMicroseconds(20);
```

```
digitalWrite(trigpin,LOW);
```

Sends a 20  $\mu$ s pulse to activate the ultrasonic sensor.

## Echo Measurement

```
duration = pulseIn(echopin, HIGH);
```

```
distance = duration * 0.034 / 2;
```

- Measures time taken for sound to return.
- Converts time  $\rightarrow$  distance (cm).
- 0.034  $\rightarrow$  speed of sound in cm per microsecond.

## Distance Result

```
if(distance <= 100)
```

```
{
```

```
    digitalWrite(BUZZER, HIGH);
```

```
    Serial.println("OBJECT IS DETECTED");
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(BUZZER, LOW);
```

```
    Serial.println("OBJECT IS NOT DETECTED");
```

```
}
```

- If an object/animal comes **within 100 cm**, buzzer turns ON.
- If no object  $\rightarrow$  buzzer OFF.

### C) PIR SENSOR – Motion Detection

```
val = digitalRead(pirPin);
```

Reads PIR sensor output:

- **HIGH** → **Motion detected**
- **LOW** → **No motion**

#### When Motion is Detected:

```
if (val == HIGH)
{
    digitalWrite(ledPin, HIGH);
    delay(1000);
    if (pirState == LOW)
    {
        Serial.println("⚠ Motion detected!");
        pirState = HIGH;
        delay(1000);
    }
}
```

- LED turns ON
- Prints "Motion detected!" only once
- Buzzer may also be ON if ultrasonic detects movement

## When Motion Stops

else

{

digitalWrite(ledPin, LOW);

if (pirState == HIGH)

{

Serial.println("Motion ended.");

pirState = LOW;

}

}

- LED OFF
- Prints "Motion ended."

## PROGRAM SUMMARY

| Component           | Action                |
|---------------------|-----------------------|
| Soil Dry            | Pump Turns ON         |
| Soil Wet            | Pump Turns OFF        |
| Object < 100 cm     | Buzzer ON             |
| No Object           | Buzzer OFF            |
| PIR Motion Detected | LED ON + Serial Alert |
| Motion Ends         | LED ON + Serial Alert |

### 3.4BILL OF MATERIALS

| Sl. No. | Component                                      | Specification / Description                 | Quantity  | Estimated Cost (INR) |
|---------|------------------------------------------------|---------------------------------------------|-----------|----------------------|
| 1       | NodeMCU ESP8266                                | Wi-Fi enabled microcontroller board         | 1         | ₹350                 |
| 2       | Soil Moisture Sensor                           | Digital/Analog output, corrosion-proof type | 1         | ₹120                 |
| 3       | Ultrasonic Sensor HC-SR04                      | Distance measuring sensor                   | 1         | ₹120                 |
| 4       | PIR Motion Sensor (HC-SR501)                   | Human/animal motion detection               | 1         | ₹120                 |
| 5       | Relay Module (1-channel)                       | 5V relay for pump control                   | 1         | ₹120                 |
| 6       | Water Pump                                     | 5V/12V DC mini pump / submersible           | 1         | ₹350                 |
| 7       | Buzzer (Active)                                | 5V buzzer for alert                         | 1         | ₹20                  |
| 8       | LED Indicator                                  | Status indication LED                       | 1         | ₹5                   |
| 9       | Jumper Wires (Male-to-Male / Female-to-Female) | For breadboard connections                  | 20–30 pcs | ₹150                 |
| 10      | Breadboard                                     | 400/830 point breadboard                    | 1         | ₹220                 |
| 11      | Rechargeable Battery                           | 12V depending on motor                      | 5         | ₹600                 |
| 12      | Battery Connector                              | For powering pump/battery                   | 1         | ₹40                  |

| Sl. No. | Component   | Specification / Description | Quantity  | Estimated Cost (INR) |
|---------|-------------|-----------------------------|-----------|----------------------|
| 13      | Water Pipes | For pump water flow         | As needed | ₹50                  |
| 14      | Total       |                             |           | ₹2265                |

## CHAPTER 4

### RESULTS AND DISCUSSION

The smart agriculture monitoring and farm protection system that was made integrates sensing soil moisture, monitoring temperature and humidity, detecting intruders, video surveillance, and wireless communication into one effective model. The ESP32-CAM made it possible for farmers to view their fields from anywhere by allowing real-time visual surveillance and remote access through the Telegram app. The autonomous irrigation system that used a soil moisture sensor only gave water when it was needed. This cut down on waste and helped keep resources available for longer. The PIR sensor-based intrusion detection system protected crops from animals and people who shouldn't be there, which lowered the chance of losing crops. The bell quickly alerted others around, and the LED display made it easy to keep an eye on things without having to rely only on a cell phone connection. This system is good for farming in rural and remote areas because it has a rechargeable battery that can also be charged by solar power.



In general, the technology was cost-effective, easy to use, and useful for real-world farming. The study shows how IoT-based automation may help farmers use modern and sustainable farming methods by reducing the need for human labour, stopping crop loss, improving water management, and allowing for remote monitoring. The system works well, but there are a number of changes that may be made to make it work better and do more. Using cloud storage to save and analyze long-term sensor data and image logs is one way to make things better. This can help farmers learn more about seasonal patterns and make better choices. In the future, machine learning algorithms might also be used to figure out how much water is needed based on changes in the environment. This would make the irrigation system completely automated, so there would be no need for threshold management.

## CHAPTER 5

### CONCLUSION

The Integrated Temperature-Based Irrigation and Animal Intrusion Prevention System using ESP8266 successfully demonstrates how a low-cost IoT approach can automate essential agricultural functions. The system accurately monitors soil moisture and controls irrigation without human intervention, ensuring optimal watering and preventing both over-irrigation and water wastage. The integration of ultrasonic and PIR sensors provides reliable detection of animals or intruders, activating a buzzer alert to protect the field from potential crop damage.

Through real-time monitoring and efficient decision-making, the system enhances farm productivity, supports sustainable water management, and improves crop security. The ESP8266 microcontroller proved to be effective in handling simultaneous sensor inputs and control outputs with stable performance. Overall, the proposed system provides a practical, economical, and scalable solution that addresses key challenges faced by modern agriculture.

### FUTURE SCOPE

Although the system performs well in real-world conditions, several enhancements can further improve its capability, reliability, and applicability:

#### 1. Integration with IoT Platforms

- Add cloud connectivity (Blynk, Firebase, Thingspeak) for remote monitoring.
- View soil moisture, intrusion alerts, pump status on a mobile app.

#### 2. Solar Power Implementation

- Integrating a solar panel can make the system fully energy-independent.
- Ideal for remote agricultural lands without electricity.

#### 3. AI-Based Animal Recognition

- Camera module + AI (YOLO, TensorFlow Lite) to identify specific animals.
- Reduces false alarms triggered by wind or small objects.

#### 4. SMS/Call Alert System

- GSM module (SIM800L) can notify farmers immediately during intrusion.

- Useful where Wi-Fi is unavailable.

### **5. Smart Irrigation Algorithms**

- Implement temperature + humidity + evapotranspiration (ET) formulas.
- More precise irrigation scheduling based on crop and climate.

### **6. Automatic Water Level Monitoring**

- Add a water tank level sensor to prevent pump dry-run conditions.

### **7. Rugged Outdoor Enclosure**

- Waterproof casing for long-term outdoor deployment in rain and heat.

### **8. Multi-Sensor Node Network**

- Several irrigation nodes communicating together for larger fields.

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## DATASHEETS OF COMPONENTS USED

### 1. NodeMCU ESP8266 Development Board – Datasheet

#### Description:

A Wi-Fi-enabled microcontroller based on ESP8266EX used for IoT applications.

#### Key Specifications:

- Microcontroller: **ESP8266EX**
- Operating Voltage: **3.3V**
- Input Voltage (USB): **5V**
- Flash Memory: **4MB**
- Digital I/O Pins: **11**
- ADC Pins: **1 (10-bit)**
- Clock Speed: **80/160 MHz**
- Wi-Fi: 802.11 b/g/n support
- Communication: UART, SPI, I2C

### 2. Soil Moisture Sensor (FC-28) – Datasheet

#### Description:

Measures the water content in soil using electrical conductivity.

#### Key Specifications:

- Input Voltage: **3.3V–5V**
- Output:
  - **Analog Output (AO)**
  - **Digital Output (DO)** via comparator
- Interface Board: LM393 comparator
- Corrosion-prone unless using corrosion-resistant version
- Output HIGH → Soil is WET
- Output LOW → Soil is DRY

### 3. Ultrasonic Sensor HC-SR04 – Datasheet

**Description:**

Measures distance using ultrasonic wave reflection.

**Key Specifications:**

- Operating Voltage: **5V**
- Working Frequency: **40 kHz**
- Measuring Range: **2 cm – 400 cm**
- Accuracy: **±3 mm**
- Trigger Input: 10  $\mu$ s pulse
- Echo Output Pulse: proportional to distance
- Quiescent Current: **< 2 mA**

**Distance Formula:**

$$\text{Distance (cm)} = \frac{\text{Time } (\mu\text{s}) \times 0.034}{2}$$

**4. PIR Motion Sensor (HC-SR501) – Datasheet**

**Description:**

Detects motion of human/animal using infrared radiation.

**Key Specifications:**

- Operating Voltage: **5V–20V**
- Detection Range: **3–7 meters**
- Output Voltage: **HIGH (3.3V)** when motion detected
- Trigger Modes: **L** (single) / **H** (retrigger)
- Delay Time: Adjustable (3–300 sec)
- Sensitivity: Adjustable
- Operating Current: **50  $\mu$ A**

**5. Relay Module (5V, 1-Channel) – Datasheet**

**Description:**

Used to control AC/DC loads like motors and pumps.

**Key Specifications:**

- Operating Voltage: **5V**
- Control Signal: **3.3V compatible**
- Switching Voltage: **AC 250V / 10A**
- DC Load: **30V / 10A**
- Relay Type: **SRD-05VDC-SL-C**
- Optocoupler Isolation: In many models

**Pinout:**

- **IN** – Signal control
- **VCC** – +5V
- **GND** – Ground
- **COM** – Common
- **NO** – Normally Open
- **NC** – Normally Closed

**6. Buzzer (Active Buzzer) – Datasheet**

**Description:**

Generates beep sound when powered.

**Key Specifications:**

- Operating Voltage: **3.3V–5V**
- Current: **20–30 mA**
- Type: Active (self-oscillating)
- Frequency: **2–4 kHz**

**7. Water Pump (5V/12V Mini Pump) – Datasheet**

**Description:**

Used for irrigation water flow.

**Key Specifications (5V pump):**

- Rated Voltage: **5V**
- Current: **100–300 mA**

- Flow Rate: **80–120 L/H**
- Lift Distance: **0.5–1 meter**

**Specifications (12V pump):**

- Rated Voltage: **12V**
- Current: **0.5–1A**
- Flow Rate: **200–300 L/H**
- Lift Distance: **2–3 meters**

**8. LED Indicator – Datasheet**

**Description:**

Standard indicator LED.

**Key Specifications:**

- Operating Voltage: **2V–2.2V**
- Recommended Current: **10–20 mA**
- Forward Current: 20 mA
- Requires **220Ω resistor** in series